THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 8: **Introduction to Trigonometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $\tan\theta = \frac{12}{5}$, then $\frac{\sin\theta + \cos\theta}{\cos\theta} =$ 1

(a) $\frac{12}{5}$ **(b)** $\frac{17}{5}$ **(c)** $\frac{7}{5}$ **(d)** $\frac{5}{17}$

2. If $\sin 35^\circ = \cos B$ for an acute angle B, then $B =$ 1

(a) 145° **(b)** 35° **(c)** 45° **(d)** 55°

3. The value of $\cos^2 45^\circ + \tan^2 45^\circ$ is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{3}{2}$ **(c)** 3 **(d)** $\frac{5}{2}$

4. If $\sin\theta = \frac{15}{17}$, then $\sec\theta + \tan\theta =$ 1

(a) $\frac{17}{8}$ **(b)** 4 **(c)** $\frac{1}{4}$ **(d)** $\frac{32}{15}$

5. If $\cos\theta = \frac{8}{17}$, then $\csc\theta - \cot\theta =$ 1

(a) $\frac{5}{3}$ **(b)** $\frac{3}{5}$ **(c)** $\frac{17}{15}$ **(d)** $\frac{1}{4}$

6. $\frac{\sin 48^\circ}{\cos 42^\circ} + \frac{\cos 55^\circ}{\sin 35^\circ} =$ 1

(a) 1 **(b)** 4 **(c)** 0 **(d)** 2

7. If $\tan\theta = \frac{15}{8}$, then $\frac{\sin\theta + \cos\theta}{\cos\theta} =$ 1

(a) $\frac{7}{8}$ **(b)** $\frac{15}{8}$ **(c)** $\frac{23}{8}$ **(d)** $\frac{8}{23}$

8. If $\sin 48^\circ = \cos B$ for an acute angle B, then $B =$ 1

(a) 48° **(b)** 42° **(c)** 132° **(d)** 45°

9. The value of $3\tan^2 30^\circ + \cos^2 60^\circ$ is: 1

(a) $\frac{5}{4}$ **(b)** $\frac{5}{2}$ **(c)** $\frac{9}{4}$ **(d)** 1

10. If $\sin\theta = \frac{24}{25}$, then $\sec\theta + \tan\theta =$ 1

(a) $\frac{25}{7}$ **(b)** 7 **(c)** $\frac{49}{24}$ **(d)** $\frac{1}{7}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. If $\tan(3x^\circ) = \sqrt{3}$, find the value of x (an acute angle). 2

12. Evaluate: $2\tan^2 60^\circ - \sin^2 30^\circ$. 2

13. If $\tan A = \frac{24}{7}$, find $\frac{\sin A + \cos A}{\sin A - \cos A}$. 2

14. If $\cos(2x + 10^\circ) = \frac{1}{2}$, find the value of x (an acute angle). 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. If $\cos(3x^\circ) = \frac{\sqrt{3}}{2}$, find the value of x (an acute angle). 3

16. If $\tan\theta = \dfrac{21}{20}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 3

17. Prove that: $\cos^2 A(1 + \tan^2 A) = 1$. 3

OR

If $\tan(2x^\circ) = \frac{1}{\sqrt{3}}$, find the value of x (an acute angle). 3

18. If $\tan\theta = \dfrac{40}{9}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 3

OR

Prove that: $\sin^2 A \cot^2 A + \cos^2 A \tan^2 A = 1$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that: $\dfrac{1 + \tan^2 A}{1 + \cot^2 A} = \left(\dfrac{1 - \tan A}{1 - \cot A}\right)^2$. 5

OR

If $\tan A = \dfrac{21}{20}$, evaluate $\dfrac{5\sin A + 3\cos A}{1\sin A + 1\cos A}$. 5

20. Prove that: $\sin^4\theta - \cos^4\theta = 2\sin^2\theta - 1$. 5

OR

If $\tan A = \dfrac{40}{9}$, evaluate $\dfrac{4\sin A + 7\cos A}{2\sin A + 1\cos A}$. 5

END OF QUESTION PAPER Total: 40 Marks

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